

Concord Lite Zone A Supplement

Target AG2 Workflow Under Test

Hackathon build companion

1. Why Zone A Exists

Zone A is the **target AG2 workflow under test**. Concord Lite cannot diagnose anything unless this workflow exists first. Zone A creates the run trace, tool events, ContextVariables snapshots, handoff path, final output, and contract violations that Zone B analyzes.

Clean boundary

- Zone A solves the original user task.
- Zone B inspects Zone A and turns the run into contract violations, repairs, and regression tests.
- For the hackathon, Zone A should be small, deterministic, and intentionally flawed in controlled ways.

Recommended demo scenario

Use a **Literature Review Assistant Run #041**. This gives us a scientific research angle while still being easy to explain. The user asks a research question. The target AG2 workflow gathers evidence, critiques it, verifies claims, writes a mini literature memo, and then tries to save or export the report.

Why literature review is the best Zone A demo

- Fits the scientific research prize if needed.
- Tavily is naturally useful for external evidence search.
- Daytona can validate generated tests or run a small citation/evidence checker.
- Missing evidence is easy for judges to understand.
- Unsupported claims feel dangerous but not legally or medically risky.

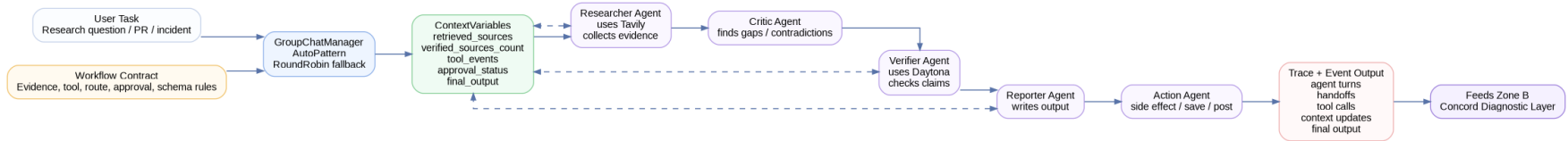


Figure 1: Zone A architecture

2. Zone A Architecture and Agent Responsibilities

Target AG2 components

Component	Purpose in Zone A	Required for demo
GroupChatManager / AutoPattern	Runs the target workflow and decides next speaker	Yes
RoundRobin fallback	Keeps the demo predictable if AutoPattern behaves unpredictably	Yes
ContextVariables	Shared state for sources, verified count, tool events, approval status, final output	Yes
Handoffs	Route from Researcher to Critic to Verifier to Reporter to ActionAgent	Yes
Researcher Agent	Uses Tavily or fixture search results to collect evidence	Yes
Critic Agent	Finds gaps, unsupported claims, or contradictions	Yes
Verifier Agent	Checks whether claims have enough support and can call Daytona for validation	Yes
Reporter Agent	Produces the final literature memo or target output	Yes
Action Agent	Represents a side effect: save report, post comment, export memo, update ticket	Yes, but mocked
UserProxyAgent / HumanGate	The approval step Zone A should have before ActionAgent	Optional in broken run, required as expected contract

Agent inputs and outputs

Agent	Input	Processing	Output / state written
Researcher Agent	User task, research question, Tavily query	Searches or loads evidence, summarizes sources	retrieved_sources[], tool_events[]
Critic Agent	retrieved_sources, draft claims	Flags weak evidence and missing support	critique_notes[], risk_flags[]
Verifier Agent	critique_notes, retrieved_sources	Checks if claims are supported and optionally runs Daytona validator	verified_sources_count, verification_result
Reporter Agent	verified evidence, critique notes	Writes final output	final_output
Action Agent	final_output, approval_status	Performs mocked side effect	action_event

ContextVariables for Zone A

Variable	Type	Who writes it	Why Concord needs it
retrieved_sources	list	Researcher	Evidence contract basis
verified_sources_count	number	Verifier	Reporter should not run if this is zero
tool_events	list	Researcher, Verifier	ToolTruth check

Variable	Type	Who writes it	Why Concord needs it
approval_status	string	UserProxyAgent	ActionAgent must require approval
handoff_path	list	GroupChatManager	Routing contract check
final_output	string/object	Reporter	Schema and evidence checks

3. The Broken Run We Intentionally Build

The Zone A run must be good enough to look real and broken enough for Concord to diagnose.

User task

Create a concise literature review memo on whether multi-agent systems improve reliability in research workflows. Include claims, evidence, and citations.

Expected contract rules

Contract	Rule
Evidence contract	Reporter may write final answer only when <code>verified_sources_count > 0</code>
Tool contract	Any claim saying “verified”, “searched”, or “checked” requires a matching <code>tool_event</code>
Routing contract	Reporter must run after Verifier, not directly after Critic
Approval contract	ActionAgent requires <code>approval_status == approved</code>
Schema contract	Final output must include <code>summary</code> , <code>claims[]</code> , <code>citations[]</code> , <code>risks[]</code> , <code>next_steps[]</code>

Intentional failures in demo trace

Failure	How we create it	What Concord will catch
Missing evidence	Verifier forgets to update <code>verified_sources_count</code>	Reporter produced final output with <code>verified_sources_count = 0</code>
Tool claim without tool event	Verifier says it checked sources but no Daytona/Tavily event is recorded	Tool contract violation
Skipped verifier or weak verifier	Handoff path shows Critic -> Reporter or Verifier with no successful tool event	Routing and evidence failure
Side effect without approval	ActionAgent receives final output while <code>approval_status</code> is pending	Approval contract violation

Trace flow

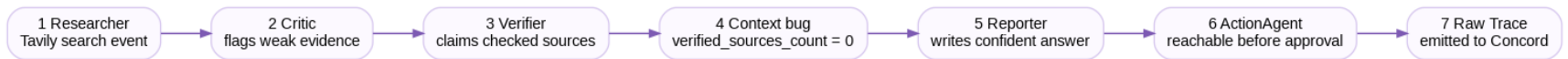


Figure 2: Zone A trace flow

Minimal raw trace event shape

```
{  
  "step": 4,  
  "agent": "VerifierAgent",  
}
```

```
"type": "agent_turn",
"content": "I verified the key claims and the evidence is sufficient.",
"tool_call_id": null,
"context_delta": {
  "verified_sources_count": 0
},
"handoff_to": "ReporterAgent"
}
```

4. What A and D Must Build for Zone A

File structure

```
zone-a/  
  agents.py  
  workflow.py  
  context.py  
  demo_trace.json  
  workflow_contract.json  
  fixtures/  
    tavily_results.json  
    papers.json  
    broken_final_output.json  
  tools/  
    tavily_search.py  
    daytona_validator.py  
  tests/  
    test_zone_a_fixture.py
```

Build tasks

Owner	Task	Done when
A	Define Zone A agents and prompts	Researcher, Critic, Verifier, Reporter, ActionAgent can run or be simulated
A	Create GroupChat / AutoPattern setup	Workflow can produce ordered trace or deterministic fixture
A	Add RoundRobin fallback	Demo can run predictably if AutoPattern fails
D	Define workflow_contract.json	Evidence, tool, routing, approval, and schema rules exist as JSON
D	Create demo_trace.json	Contains at least 10 events and 4 intentional violations
D	Add Tavily fixture or live search	UI can show Tavily evidence source or captured response
D	Add Daytona validation path	Generated test can run or simulate in Daytona
P	Render Zone A graph in frontend	Judges can see which agent failed and where
N	Write sample research question and polished trace copy	Demo feels real, not random placeholder text

Minimal Zone A API outputs

Endpoint	Output
/api/zone-a/run	run_id, workflow_name, status, final_output

Endpoint	Output
/api/zone-a/trace	events[], context_snapshots[], handoff_path[], tool_events[]
/api/zone-a/contract	workflow_contract.json
/api/zone-a/summary	human-readable summary for UI cards

What can be mocked

- Actual literature search can use Tavily fixture if network fails.
- ActionAgent side effect must be mocked.
- AutoPattern can be replaced by fixed event sequence in emergency.
- Daytona can run a tiny local-looking validation payload if sandbox setup fails, but UI should clearly show the intended Daytona role.

5. How Zone A Connects to Zone B

Handoff between layers

Zone A emits these artifacts:

1. **Raw AG2 trace:** every agent turn, handoff, tool call, context update, and final output.
2. **Workflow contract:** expected rules for evidence, tools, routing, approval, and schema.
3. **Context snapshots:** state before and after key agent turns.
4. **Tool events:** Tavily search events, Daytona validation events, or captured fixtures.
5. **Final output:** the artifact Concord checks for contract violations.

Zone B consumes those artifacts and produces:

1. Contract Violation Report
2. Failed agent and failed step as supporting metadata
3. Repair patch mapped to AG2 primitive
4. Generated regression test
5. Human-approved rerun-ready recommendation

Demo order

Demo moment	What appears on screen	Why it matters
1	Literature Review Assistant Run #041	Concrete target workflow
2	Researcher -> Critic -> Verifier -> Reporter -> ActionAgent	Visible multi-agent system
3	Trace shows Verifier claims verification but no matching tool event	Tool contract violation is obvious
4	ContextVariables show verified_sources_count = 0	Evidence contract violation is obvious
5	ActionAgent path appears before approval	Approval contract violation is obvious
6	Concord diagnoses and repairs	Zone B payoff

Acceptance criteria for Zone A

- The target workflow can be shown visually.
- The trace has at least 10 events.
- The trace contains at least 4 controlled failures.
- The trace and contract are valid JSON.
- Zone B can consume Zone A output without manual editing.
- Tavily, Daytona, AG2, V0, and Gemini roles are clear to the team.

Final note

Zone A does not need to be a production-grade research assistant. It must be a believable, AG2-native workflow that generates a high-quality trace with controlled failures. Zone B is where the product value lives, but Zone A is what makes that value visible.